3.2.5 During the cell cycle, genetic information is copied and passed to genetically identical daughter cells.

Replication of DNA	The semi-conservative replication of DNA in terms of • breaking of hydrogen bonds between polynucleotide strands • attraction of new DNA nucleotides to exposed bases and base pairing • role of DNA helicase and of DNA polymerase. **Candidates should be able to** analyse, interpret and evaluate data concerning early experimental work relating to the role and importance of DNA.
Mitosis	During mitosis, the parent cell divides to produce two daughter cells, each containing an exact copy of the DNA of the parent cell. Mitosis increases the cell number in this way in growth and tissue repair. **Candidates should be able to** name and explain the events occurring during each stage of mitosis. They should be able to recognise the stages from drawings and photographs.
Cell cycle	Mitosis and the cell cycle. DNA is replicated and this takes place during interphase. **Candidates should be able to** relate their understanding of the cell cycle to cancer and its treatment.